

Derivative Pricing

(5. Change of Numeraire)



Definitions

Consider a market with riskless money market account $\mathbf{B}$, a risky stock $\mathbf{S}$. The portfolio process (ϕ_t, ψ_t) holds ϕ_t amount of $\mathbf{S}$ and ψ_t amount of $\mathbf{B}$ at any time t . The value of the portfolio at time t is V_t . The portfolio is **self-financing** if

$$dV_t = \phi_t dS_t + \psi_t dB_t$$

that is change in portfolio is purely governed from profits or losses of the underlying security. No money is added or removed. Now if there is a contingent claim X defined at T , a **replicating portfolio** strategy is a self-financing portfolio (ϕ_t, ψ_t) such that

$$V_T = \phi_T S_T + \psi_T B_T = X$$

Black Scholes World



Let's consider a European call option whose claim X at time T is defined as $X = (S_T - K)^+$. The dynamics of stock and bond in the physical measure $\mathbb{P}$ are given as

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

$$dB_t = rB_t dt$$

The value of the contingent claim at any time t is given as

$$V_t = B_t \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_t \right)$$

There is a 5-step method to arrive at the result.

Step 1 – Find an equivalent martingale measure

Establish a unique martingale measure $\mathbb{Q}$ under which discounted stock price $Z_t = \frac{S_t}{B_t}$ is a martingale.

Under $\mathbb{P}$ the processes are given as

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right] \quad , \quad B_t = \exp(rt)$$

$$Z_t = S_0 \exp \left[\left(\mu - r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right]$$

Using C.M.G., we can find a measure $\mathbb{Q}$ such that Z_t is a martingale.

$$Z_t = S_0 \exp \left[\left(-\frac{1}{2} \sigma^2 \right) t + \sigma \tilde{W}_t \right] \quad \text{where } \tilde{W}_t \text{ is a } \mathbb{Q} - \text{brownian}$$

Step 2 – Propose the price of the derivative

We propose the fair price of the derivative is

$$V_t = B_t \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_t \right)$$

or

$$V_t = \exp(-r(T-t)) \mathbb{E}^{\mathbb{Q}}(X | \mathcal{F}_t)$$

Step 3 – Discounted claim process is a martingale

Define E_t as the discounted claim process conditional on filtration $(\mathcal{F}_t)$

$$E_t = \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{\underline{B_T}} \middle| \mathcal{F}_t \right)$$

E_t is a martingale under $\mathbb{Q}$ (Tower law).

$$\begin{aligned} E_T &= \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_T \right) \\ &= \frac{X}{B_T} \end{aligned}$$

Step 4 – Martingale Representation Theorem

Martingale Representation Theorem :

The two $\mathbb{Q}$ martingales (Z_t, E_t) are interlinked via a previsible process ϕ_t which is the amount of stock in the replicating portfolio.

$$E_t = E_0 + \int_0^t \phi_s dZ_s$$

Or

$$dE_t = \phi_t dZ_t$$

Step 5 – Replicating Portfolio

We need to establish a replicating portfolio $\pi_t = \{\phi_t, \psi_t\}$ such that it matches the price of the contingent claim at time T . So as per **no-arbitrage**, the value of the replicating portfolio at any time should be equal to the price of the claim.

We already know what ϕ_t needs to be from martingale representation theorem. The amount of money market security at time t in the portfolio is $\psi_t = E_t - \phi_t Z_t$

Portfolio value at time t

$$\underline{V_t} = \phi_t S_t + \psi_t B_t = B_t \left(\phi_t \frac{S_t}{B_t} + \psi_t \right) = \underline{B_t E_t}$$

This portfolio is self-financing. (Show that $dV_t = \phi_t dS_t + \psi_t dB_t$)

We also see that, $\underline{V_T} = \underline{B_T E_T} = \underline{X}$ +

Hence, V_t must be the price of the claim at t , or $\underline{V_t} = \underline{B_t} \mathbb{E}^{\mathbb{Q}} \left(\frac{\underline{X}}{\underline{B_T}} \middle| \mathcal{F}_t \right)$

Pricing Formula

The price of a derivative is given as expectation of discounted payoff under risk-neutral density

$$\frac{B_t}{B_T} = e^{-r(T-t)}$$

$$\begin{aligned} \rightarrow V_t &= B_t \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_t \right) \\ V_t &= \mathbb{E}^{\mathbb{Q}} (X \exp(-r(T-t)) \mid \mathcal{F}_t) \end{aligned}$$

At time 0, the price is given as

$$V(s, t)$$

$$\begin{aligned} V_0 &= B_0 \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_0 \right) \\ V_0 &= \mathbb{E}^{\mathbb{Q}} (X \exp(-rT) \mid \mathcal{F}_0) \end{aligned}$$

Fundamental Theorem of Asset Pricing (FTAP) :

- 1st FTAP - No arbitrage portfolios require existence of at least one risk-neutral measure.
- 2nd FTAP - A complete market with no arbitrage implies the risk-neutral measure is unique.

The virtuous cycle of pricing

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Propose a portfolio process $\pi_t = \{\phi_t, \psi_t\}$ where ϕ_t is the amount of stock. The amount ϕ_t is known pre-visibly as per martingale representation theorem.

$$\underline{E_t} = E_0 + \int_0^t \phi_s dZ_s$$

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Show that the portfolio is self-financing and it matches the value of the claim X at T . So, value of the portfolio at t must also be same as the derivative at t

$$dV_t = \phi_t dS_t + \psi_t dB_t$$

$$V_T = B_T E_T = X$$

$$V_t = B_t \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_t \right) \checkmark$$

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Discounted claim conditional on filtration $\mathcal{F}_t$ is a $\mathbb{Q}$ -martingale (Tower Law)

$$\underline{E_t} = \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_t \right) \checkmark$$

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Propose the price of contingent claim X defined at T

$$V_t = B_t \mathbb{E}^{\mathbb{Q}} \left(\frac{X}{B_T} \middle| \mathcal{F}_t \right)$$

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Find an equivalent martingale measure $\mathbb{Q}$, where discounted stock price $\underline{Z_t} = \frac{S_t}{B_t}$ is a martingale using C.M.G

$$Z_t = S_0 \exp \left[\left(-\frac{1}{2} \sigma^2 \right) t + \sigma \underline{\tilde{W}_t} \right] \quad \underline{\tilde{W}_t} \text{ is } \mathbb{Q} - \text{brownian}$$

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Black Scholes Dynamics under $\mathbb{P}$ ✓

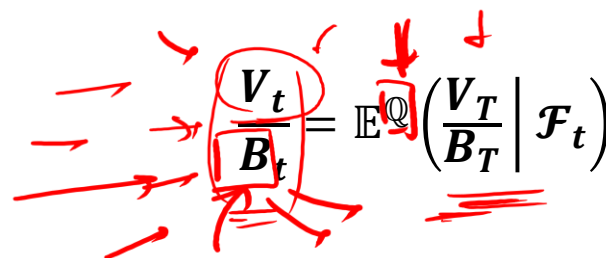
$$dS_t = \underline{\mu} S_t dt + \sigma S_t dW_t$$

$$dB_t = r B_t dt \rightarrow$$



The Concept of Numeraire

If you inspect the formula for derivative, we can see that the term under the expectation is asset price discounted by the cash bond and it's a martingale. The cash bond serves as a reference asset. Intuitively speaking, in the risk-neutral world, everything grows at risk-free rate matching the drift of the reference asset. In other words, there is no relative drift w.r.t the reference asset. This reference asset is called a **numeraire**. It's like a unit of currency against which every asset value is measured.



$$\frac{V_t}{B_t} = \mathbb{E}^{\mathbb{Q}} \left(\frac{V_T}{B_T} \mid \mathcal{F}_t \right)$$

The measure $\mathbb{Q}$ enforces a drift of r for every asset which matches the drift of the reference asset. This makes the ratio of any asset to the reference asset driftless under $\mathbb{Q}$

X/s

A **numeraire** is any positive non-dividend paying tradeable asset. So, you can think of stocks, bonds, money market, commodities as examples of numeraires. Is FX rate a numeraire?

Examples of Numeraires

Consider X_t to be a tradeable-asset. Under different measures and the corresponding numeraires, the dynamics of X_t will be different. Some choices make the math easy depending on the problem.

- **Risk-Neutral** measure is associated with **money market** $B(t)$ numeraire.

$$\frac{dX_t}{X_t} = rdt + \sigma dW_t^{\mathbb{Q}} \Rightarrow \mathbb{E}^{\mathbb{Q}}\left(\frac{X_t}{B_t} \middle| \mathcal{F}_0\right) = \frac{X_0}{B_0}$$

- **T-Forward** $\mathbb{Q}^T$ measure is associated with ZCB, $P(t, T)$ as the numeraire (more on this later!)

$$\frac{dX_t}{X_t} = \mu^T dt + \sigma dW_t^{\mathbb{T}} \Rightarrow \mathbb{E}^{\mathbb{T}}\left(\frac{X_t}{P(t, T)} \middle| \mathcal{F}_0\right) = \frac{X_0}{P(0, T)}$$

- **Stock measure** $\mathbb{Q}^S$ is associated with **stock** $S(t)$ as the numeraire

$$\frac{dX_t}{X_t} = \mu^S dt + \sigma dW_t^{\mathbb{S}} \Rightarrow \mathbb{E}^{\mathbb{S}}\left(\frac{X_t}{S(t)} \middle| \mathcal{F}_0\right) = \frac{X_0}{S(0)}$$

Change of Numeraire

Assume there exists a numeraire N and a probability measure $\mathbb{Q}^N$, such that the price of any traded asset X relative to N is a martingale under $\mathbb{Q}^N$, i.e.,

$$\rightarrow \frac{X_t}{N_t} = \mathbb{E}^N \left(\frac{X_T}{N_T} \middle| \mathcal{F}_t \right) \quad 0 \leq t \leq T$$

Let U be an arbitrary numeraire. Then there exists a probability measure $\mathbb{Q}^U$, equivalent to $\mathbb{Q}^N$, such that the price X normalized by U is a martingale under $\mathbb{Q}^U$, i.e.,

$$\rightarrow \frac{X_t}{U_t} = \mathbb{E}^U \left(\frac{X_T}{U_T} \middle| \mathcal{F}_t \right) \quad 0 \leq t \leq T$$

The Radon-Nikodym derivative associated with the change of measure from $\mathbb{Q}^N$ to $\mathbb{Q}^U$ is given as

$$\rightarrow \frac{d\mathbb{Q}^U}{d\mathbb{Q}^N} = \frac{U_T N_t}{U_t N_T} \quad (\text{Prove it!})$$

Change of Numeraire – Application (I)

Assume Black-Scholes dynamics under risk-neutral world $\mathbb{Q}$

$$\begin{aligned} \rightarrow \quad \underline{dS_t} &= \underline{rS_t dt} + \underline{\sigma S_t d\tilde{W}_t} & \tilde{W}_t \text{ is } \underline{\mathbb{Q}} - \text{brownian} \\ \underline{dB_t} &= \underline{rB_t dt} \end{aligned}$$

Find the price of the derivative which has a pay-off

$$X = \max [\underline{S^2(T)} - \underline{S(T)K}, 0]$$

Change of Numeraire – Application (II)

We know the pricing formula in the risk-neutral world is given as

$$\rightarrow V_0 = B_0 \mathbb{E}^{\mathbb{Q}} \left(\frac{\max [S^2(T) - S(T)K, 0]}{B_T} \middle| \mathcal{F}_0 \right)$$

Handwritten notes: A red circle around e^{2y} and a red arrow pointing to the $\mathbb{Q}$ measure in the expectation.

This expression looks formidable. Can we use change of numeraire technique to make it simpler. Let's try to pivot to stock measure $\mathbb{Q}^S$.

The change of measure between $\mathbb{Q}$ and $\mathbb{Q}^S$ is given by RN

$$\left\{ \frac{d\mathbb{Q}^S}{d\mathbb{Q}} = \frac{S_T B_0}{S_0 B_T} \right\}$$

Handwritten notes: A red circle around $d\mathbb{Q}^S$, a red circle around $S_T B_0$, and a red arrow pointing from the fraction to a box containing a right-pointing arrow.

Change of Numeraire – Application (III)

$$\begin{aligned}
 \underline{V_0} &= \underline{B_0} \mathbb{E}^{\mathbb{Q}} \left(\frac{\max [S^2(T) - S(T)K, 0]}{B_T} \middle| \mathcal{F}_0 \right) \\
 &= B_0 \int_{\Omega} \frac{\max [S^2(T) - S(T)K, 0]}{B_T} d\mathbb{Q} \\
 &= B_0 \int_{\Omega} \frac{\max [S^2(T) - S(T)K, 0]}{B_T} \left(\frac{S_0 B_T}{S_T B_0} \right) d\mathbb{Q}^S \\
 &= \cancel{B_0} \mathbb{E}^S \left(\frac{\max [S^2(T) - S(T)K, 0]}{B_T} \left(\frac{S_0 \cancel{B_T}}{\cancel{S_T} B_0} \right) \middle| \mathcal{F}_0 \right) \\
 &= \underline{S_0} \mathbb{E}^S (\max [S(T) - K, 0] | \mathcal{F}_0) \quad (e^{-rT})
 \end{aligned}$$

$$\begin{aligned}
 x &\sim f(x), \quad dp = f(x) dx \\
 \mathbb{E}(x) &= \int x f(x) dx = \int x dp
 \end{aligned}$$

The solution to this expectation is known from Black-Scholes formula. We now only need to know the dynamics of stock price under the stock measure $\mathbb{Q}^S$. How to derive this ?

Change of Numeraire – Application (IV)

Remember all assets discounted by stock price are martingales under the stock measure. Therefore, the money market account discounted by stock price would also be a martingale under $\mathbb{Q}^S$. We know from Ito calculus

$$\begin{aligned} ds/s &= rdt + \sigma d\tilde{w} \\ dB/B &= rdt \\ \mathbb{Q} &\rightarrow \mathbb{Q}^S \end{aligned}$$

$$\begin{aligned} d\left(\frac{X_t}{Y_t}\right) &= \frac{X_t}{Y_t} \left(\frac{dX_t}{X_t} - \frac{dY_t}{Y_t} - \left(\frac{dX_t}{X_t} \right) \left(\frac{dY_t}{Y_t} \right) + \left(\frac{dY_t}{Y_t} \right)^2 \right) \\ d\left(\frac{B_t}{S_t}\right) &= \left(\frac{B_t}{S_t} \right) \left(\frac{dB_t}{B_t} - \frac{dS_t}{S_t} - \left(\frac{dB_t}{B_t} \right) \left(\frac{dS_t}{S_t} \right) + \left(\frac{dS_t}{S_t} \right)^2 \right) \\ &\quad rdt - rdt - \sigma d\tilde{w} + \sigma^2 dt \end{aligned}$$

Under $\mathbb{Q}$, $d\left(\frac{B_t}{S_t}\right) = \left(\frac{B_t}{S_t}\right) (-\sigma d\tilde{W}_t + \sigma^2 dt)$

It has a drift. But we know under $\mathbb{Q}^S$ it needs to be driftless i.e. $d\left(\frac{B_t}{S_t}\right) = \left(\frac{B_t}{S_t}\right) (-\sigma dW_t^S)$ where W_t^S is a $\mathbb{Q}^S$ – Brownian. C.M.G gives us $d\tilde{W}_t = dW_t^S + \sigma dt$

$$\begin{aligned} \sigma d\tilde{w} &= \sigma dW^S + \sigma^2 dt \\ -\sigma d\tilde{w} + \sigma^2 dt &= -\sigma dW^S \end{aligned}$$

Change of Numeraire – Application (V)

Substituting in the dynamics of $S(t)$ we get,

$$\begin{aligned}
 dS_t &= \underline{rS_t dt} + \underline{\sigma S_t d\tilde{W}_t} \\
 &= \underline{rS_t dt} + \underline{\sigma S_t (dW_t^S + \sigma dt)} \\
 &= \underbrace{(r + \sigma^2)S_t dt}_{\text{(new drift)}} + \underline{\sigma S_t dW_t^S}
 \end{aligned}$$

Just replace the drift r with this $(r + \sigma^2)$ in BSM

$$\begin{aligned}
 \mathbb{E}^Q[e^{-rT} \cdot (S_T - K)^+] &= (BSM) \\
 \mathbb{E}^Q[(S_T - K)^+] &= (BSM) \times e^{rT} \\
 \mathbb{E}^S[(S_T - K)^+] &= \frac{(BSM \times e^{rT})}{r \rightarrow (r + \sigma^2)}
 \end{aligned}$$